



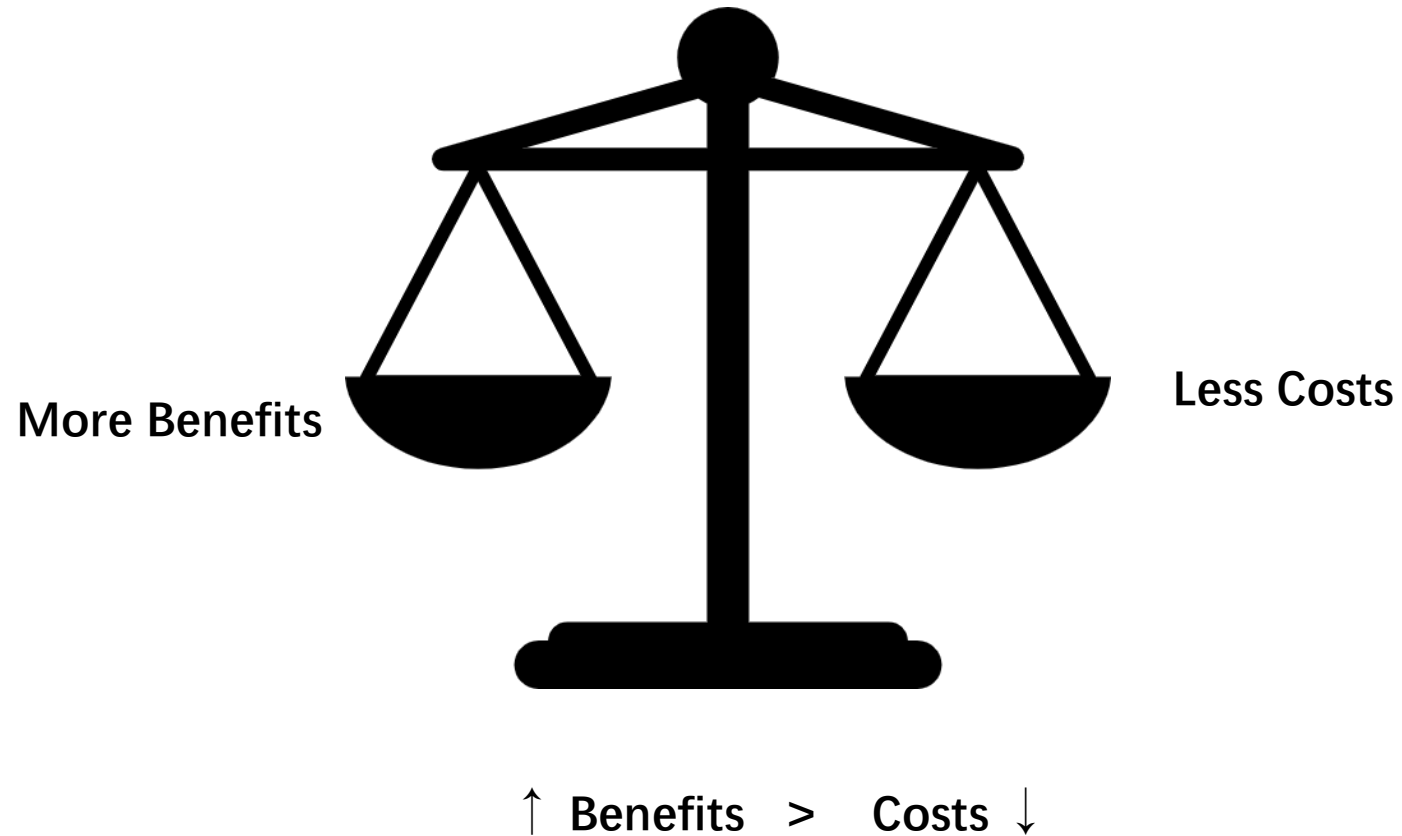
香港中文大學(深圳)  
The Chinese University of Hong Kong, Shenzhen



SCHOOL OF  
DATA SCIENCE  
數據科學學院

# Democratizing LLMs to broader applications

# A valuable LLM application should be?



# The overall picture to democratize LLMs

- **Better Benefits**

- Doing valuable applications (**LLM Applications**) to make our research more sustainable.
  - Everyone needs **healthcare**; 5%-20% GPT for healthcare for most countries
  - **One road one belt** (Arabic LLMs and multilingual LLMS)
  - **AI reshapes Education** via Agent and better interactions
- Better performance especially using open-source models (**do better Math for LLMs**)
  - There is no Golden evaluation standard for most domains, but **math**

- **Less Costs**

- Easier interaction (**multimodal interaction**)
  - Speech and multimodal interaction, plus gaming
- Computing efficiency
  - Make it faster and cheaper

# I. Which Applications

Research burns money, we have to make our research somehow application-driven



## **Criteria to select the application**

- Where people cares or has potential to earn money (maybe in the future)
- For social welfare (companies might not do this)
- Local advantages (CUHKSZ, the city Shenzhen, or broader the great bay area)

## **The chosen one for our group:**

- Healthcare
- Education
- Applications for One road and one belt.

# I. Application 1: LLMs for healthcare

## HuatuoGPT series

- 2023 May, **HuatuoGPT**: Taming Language Models to Be a Doctor. (Findings of EMNLP 2023.) 1k GitHub stars
- 2023 May, Huatuo-26M: the largest medical QA **dataset**. (Findings of NAACL 2023).
- 2023 Nov, **HuatuoGPT-II**: One-stage Training for Medical Adaptation of LLMs. (COLM 2024). 1k GitHub stars
- 2024 March, **Apollo: multilingual** medical LLMs for 6 languages
- 2024 June, **HuatuoGPT-Vision: Multimodal (visual)** Medical LLMs. (EMNLP 2024). 200, 000 HF downloading
- 2024 Oct, HuatuoGPT **deployed** in 12 local hospitals
- 2024 Nov. **Apollo 2: multilingual** medical LLMs for 50 languages. (ICLR 2025)
- 2024 Dec, **HuatuoGPT-o1**: Towards Medical **Complex Reasoning** with LLMs. (Findings of ACL 2025). 1k GitHub stars
- 2024 June, TCM-omni: **Multimodal (audio/vision/EEG) LLMs for Traditional Chinese Medicine**.

[1] Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, Jianye Hou, **Benyou Wang**. HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs. <https://arxiv.org/abs/2412.18925>

[2] Junying Chen, Ruyi Ouyang, Anningzhe Gao, Shunian Chen, Guiming Hardy Chen, Xidong Wang, Ruifei Zhang, Zhenyang Cai, Ke Ji, Guangjun Yu, Xiang Wan, **Benyou Wang**. HuatuoGPT-Vision, Towards Injecting Medical Visual Knowledge into Multimodal LLMs at Scale. <https://arxiv.org/abs/2406.19280>. EMNLP 2024

[3] Junying Chen, Xidong Wang, Anningzhe Gao#, Feng Jiang, Shunian Chen, Hongbo Zhang, Dingjie Song, Wenya Xie, Chuyi Kong, Jianquan Li, Xiang Wan, Haizhou Li, **Benyou Wang**. HuatuoGPT-II, one-stage training for medical adaption of LLMs. <https://arxiv.org/abs/2311.09774>. COLM 2024

[4] Hongbo Zhang, Junying Chen, Feng Jiang, Fei Yu, Zhihong Chen, Jianquan Li, Guiming Chen, Xiangbo Wu, Zhiyi Zhang, Qingying Xiao, Xiang Wan, **Benyou Wang**, Haizhou Li. HuatuoGPT, towards Taming Language Model to Be a Doctor. <https://arxiv.org/abs/2305.15075>. Findings of EMNLP 2023

[5] Zhihong Chen, Shizhe Diao, **Benyou Wang**, Guanbin Li, and Xiang Wan. "Towards Unifying Medical Vision-and-Language Pre-training via Soft Prompts". **ICCV 2023**

[6] Xidong Wang, Guiming Hardy Chen, Dingjie Song, Zhiyi Zhang, Zhihong Chen, Qingying Xiao, Feng Jiang, Jianquan Li, Xiang Wan, **Benyou Wang**, Haizhou Li. CMB: A Comprehensive Medical Benchmark in Chinese. **NAACL 2024**

# Impact: deployed in 12 hospitals



国内首个区域信息平台大模型智能导诊系统，实现了导诊、咨询、科普三大功能。

华佗GPT已在龙岗区**12家**医院上线使用，实现了龙岗区属公立医院全覆盖，智能导诊系统（包含导诊、客服咨询、健康咨询）已服务超过24万人次，交互次数38万，分诊准确率达95.2%。

# Roadmap to Generalist Medical AI (GMAI)



Medical Education

GMAI  
通用医疗人工智能

Preparing more data for the **patient** agents  
Collect more feedback from the **doctors**  
**Self play** between patient vs. doctor agents



[1] Junying Chen, Chi Gui, Anningzhe Gao, Ke Ji, Xidong Wang, Xiang Wan, **Benyou Wang**. CoD, Towards an Interpretable Medical Agent using Chain of Diagnosis.  
<https://arxiv.org/abs/2407.13301>

[2] Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, Jianye Hou, **Benyou Wang**. HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs.  
<https://arxiv.org/abs/2412.18925>

# I. Application 2: One Road one Belt

**AceGPT: value alignment for a new language (Arabic)**

**AceGPT 1.5: vocabulary expansion**

**AceGPT 2: native alignment**

**Apollo: multilingual medical LLMs to 6 languages**

**Apollo-2: multilingual medical LLMs to 50 languages**

[1] Efficiently Democratizing Medical LLMs for 50 Languages via a Mixture of Language Family Experts. Guorui Zheng , Xidong Wang , Juhao Liang, Nuo Chen, Yuping Zheng, **Benyou Wang\***. <https://arxiv.org/pdf/2410.10626>. submitted to **ICLR 2025** (score : 8666)

[2] Xidong Wang, Nuo Chen, Junying Chen, Yan Hu, Yidong Wang, Xiangbo Wu, Anningzhe Gao, Xiang Wan, Haizhou Li, **Benyou Wang**. Asclepius: a Lightweight Multilingual Medical LLM towards Democratizing Medical AI to 6B People. In progress.

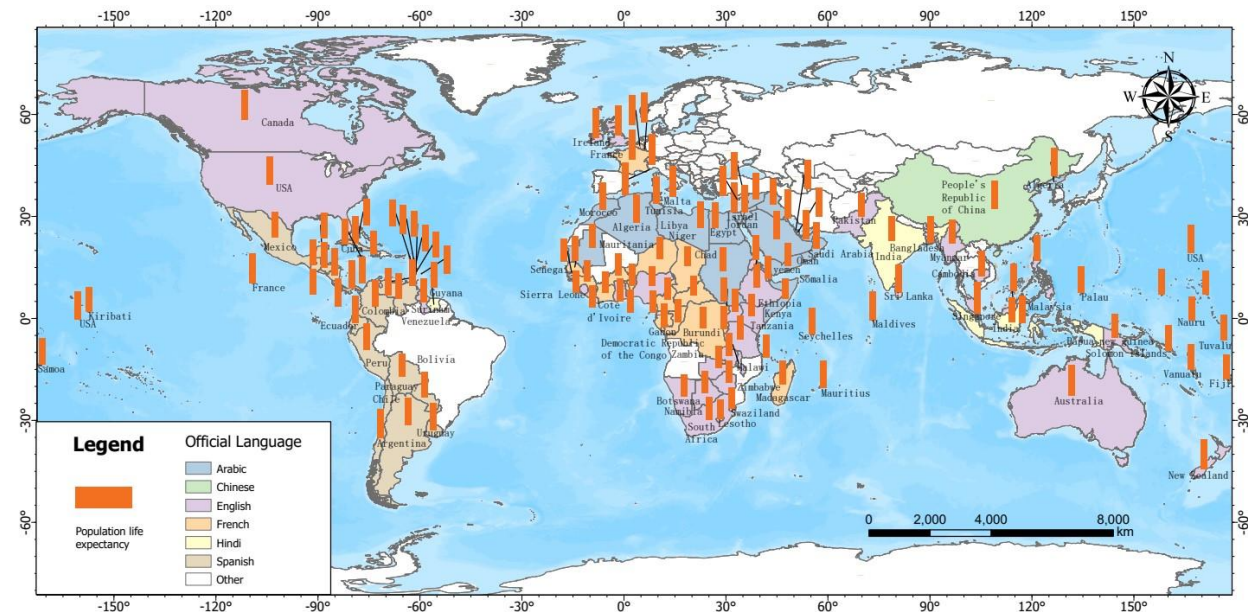
[3] Juhao Liang, Zhenyang Cai, Jianqing Zhu, Huang Huang, Kewei Zong, Bang An, Mosen Alharthi, Juncai He, Lian Zhang, Haizhou Li, **Benyou Wang**, Jinchao Xu. Alignment at Pre-training! Towards Native Alignment for Arabic **LLMs. NeurIPS 2024**

[4] Jianqing Zhu, Huang Huang, Zhihang Lin, Juhao Liang, Zhengyang Tang, Khalid Almubarak, Mosen Alharthi, Bang An, Juncai He, Xiangbo Wu, Fei Yu, Junying Chen, MA Zhuoheng, Yuhao Du, Yan Hu, He Zhang, Emad A. Alghamdi, Lian Zhang, Ruoyu Sun, Haizhou Li, Jinchao Xu, **Benyou Wang**. Second Language (Arabic) Acquisition of LLMs via Progressive Vocabulary Expansion. (**meta score of 5, recommended to ACL by AC**)

[5] Huang Huang, Fei Yu, Jianqing Zhu, Xuening Sun, Hao Cheng, Dingjie Song, Zhihong Chen, Abdulmohsen Alharthi, Bang An, Ziche Liu, Zhiyi Zhang, Junying Chen, Jianquan Li, **Benyou Wang**, Lian Zhang, Ruoyu Sun, Xiang Wan, Haizhou Li, Jinchao Xu. AceGPT, Localizing Large Language Models in Arabic. **NAACL 2024**



# I. Application 2: One Road one Belt



Democratizing Medical LLMs to 6B People and then 50 languages

[1] Efficiently Democratizing Medical LLMs for 50 Languages via a Mixture of Language Family Experts. Guorui Zheng , Xidong Wang , Juhao Liang, Nuo Chen, Yuping Zheng, **Benyou Wang\***. <https://arxiv.org/pdf/2410.10626>. submitted to **ICLR 2025** (score : 8666)

[2] Xidong Wang, Nuo Chen, Junying Chen, Yan Hu, Yidong Wang, Xiangbo Wu, Anningzhe Gao, Xiang Wan, Haizhou Li, **Benyou Wang**. Asclepius: a Lightweight Multilingual Medical LLM towards Democratizing Medical AI to 6B People. In progress.

[3] Juhao Liang, Zhenyang Cai, Jianqing Zhu, Huang Huang, Kewei Zong, Bang An, Mosen Alharthi, Juncai He, Lian Zhang, Haizhou Li, **Benyou Wang**, Jinchao Xu. Alignment at Pre-training! Towards Native Alignment for Arabic **LLMs**. **NeurIPS 2024**

[4] Jianqing Zhu, Huang Huang, Zhihang Lin, Juhao Liang, Zhengyang Tang, Khalid Almubarak, Mosen Alharthi, Bang An, Juncai He, Xiangbo Wu, Fei Yu, Junying Chen, MA Zhuoheng, Yuhao Du, Yan Hu, He Zhang, Emad A. Alghamdi, Lian Zhang, Ruoyu Sun, Haizhou Li, Jinchao Xu, **Benyou Wang**. Second Language (Arabic) Acquisition of LLMs via Progressive Vocabulary Expansion. (**meta score of 5, recommended to ACL by AC**)

[5] Huang Huang, Fei Yu, Jianqing Zhu, Xuening Sun, Hao Cheng, Dingjie Song, Zhihong Chen, Abdulmohsen Alharthi, Bang An, Ziche Liu, Zhiyi Zhang, Junying Chen, Jianquan Li, **Benyou Wang**, Lian Zhang, Ruoyu Sun, Xiang Wan, Haizhou Li, Jinchao Xu. AceGPT, Localizing Large Language Models in Arabic. **NAACL 2024**

# I. Application 2: One Road one Belt



[1] Efficiently Democratizing Medical LLMs for 50 Languages via a Mixture of Language Family Experts. Guorui Zheng , Xidong Wang , Juhao Liang, Nuo Chen, Yuping Zheng, **Benyou Wang\***. <https://arxiv.org/pdf/2410.10626>. submitted to **ICLR 2025** (score : 8666)

[2] Xidong Wang, Nuo Chen, Junying Chen, Yan Hu, Yidong Wang, Xiangbo Wu, Anningzhe Gao, Xiang Wan, Haizhou Li, **Benyou Wang**. Asclepius: a Lightweight Multilingual Medical LLM towards Democratizing Medical AI to 6B People. In progress.

[3] Juhao Liang, Zhenyang Cai, Jianqing Zhu, Huang Huang, Kewei Zong, Bang An, Mosen Alharthi, Juncal He, Lian Zhang, Haizhou Li, **Benyou Wang**, Jinchao Xu. Alignment at Pre-training! Towards Native Alignment for Arabic **LLMs. NeurIPS 2024**

[4] Jianqing Zhu, Huang Huang, Zhihang Lin, Juhao Liang, Zhengyang Tang, Khalid Almubarak, Mosen Alharthi, Bang An, Juncal He, Xiangbo Wu, Fei Yu, Junying Chen, MA Zhuoheng, Yuhao Du, Yan Hu, He Zhang, Emad A. Alghamdi, Lian Zhang, Ruoyu Sun, Haizhou Li, Jinchao Xu, **Benyou Wang**. Second Language (Arabic) Acquisition of LLMs via Progressive Vocabulary Expansion. (**meta score of 5, recommended to ACL by AC**)

[5] Huang Huang, Fei Yu, Jianqing Zhu, Xuening Sun, Hao Cheng, Dingjie Song, Zhihong Chen, Abdulmohsen Alharthi, Bang An, Ziche Liu, Zhiyi Zhang, Junying Chen, Jianquan Li, **Benyou Wang**, Lian Zhang, Ruoyu Sun, Xiang Wan, Haizhou Li, Jinchao Xu. AceGPT, Localizing Large Language Models in Arabic. **NAACL 2024**

## II. AI for Math

### OpenAI Imagines Our AI Future

#### Stages of Artificial Intelligence

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|         |                                                           |
|---------|-----------------------------------------------------------|
| Level 1 | Chatbots, AI with conversational language                 |
| Level 2 | Reasoners, human-level problem solving                    |
| Level 3 | Agents, systems that can take actions                     |
| Level 4 | Innovators, AI that can aid in invention                  |
| Level 5 | Organizations, AI that can do the work of an organization |

Source: Bloomberg reporting

👤 公众号 · 新智元

Zhengyang Tang, Xingxing Zhang, **Benyou Wang**, Furu Wei. MathScale: Scaling Instruction Tuning for Mathematical Reasoning, **ICML 2024**

Zhengyang Tang, Chenyu Huang, Xin Zheng, Shixi Hu, Zizhuo Wang, Dongdong Ge, **Benyou Wang**. ORLM: Training Large Language Models for Optimization Modeling.

<https://arxiv.org/abs/2405.17743> **Operations Research**

Xuhan Huang, Qingning Shen, Yan Hu, Anningzhe Gao, **Benyou Wang**. Mamo: a Mathematical Modeling Benchmark with Solvers. <https://arxiv.org/abs/2405.13144v1> **ACL ARR** (meta score of 4)

Fei Yu, Anningzhe Gao, **Benyou Wang**. OVM, Outcome-supervised Value Models for Planning in Mathematical Reasoning. <https://arxiv.org/abs/2311.13951>. **Findings of NAACL 2024**.

Bofei Gao, Feifan Song, Zhe Yang, Zefan Cai, Yibo Miao, Qingxiu Dong, Lei Li, Chenghao Ma, Liang Chen, Runxin Xu, Zhengyang Tang, Benyou Wang, Daoguang Zan, Shanghaoran Quan, Ge Zhang, Lei Sha, Yichang Zhang, Xuancheng Ren, Tianyu Liu, Baobao Chang . Omni-math: A universal olympiad level mathematic benchmark for large language models. <https://arxiv.org/abs/2410.07985> (ICLR 2025, 8865)

# II. AI for Math

| Rank | Model                                                       | Accuracy↑ | Parameters<br>(Billion) | Extra<br>Training<br>Data | Paper                                                                                                       | Code | Result | Year | Tags                                |
|------|-------------------------------------------------------------|-----------|-------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------|------|--------|------|-------------------------------------|
| 1    | GPT-4 DUP                                                   | 97.1      |                         | ×                         | Achieving >97% on GSM8K: Deeply Understanding the Problems Makes LLMs Better Solvers for Math Word Problems |      |        | 2024 |                                     |
| 2    | GPT-4 Code Interpreter<br>(CSV, K=5)                        | 97.0      |                         | ×                         | Solving Challenging Math Word Problems Using GPT-4 Code Interpreter with Code-based Self-Verification       |      |        | 2023 | majority voting<br>zero-shot        |
| 3    | GPT-4<br>(Model Selection, SC K=15)                         | 96.8      |                         | ×                         | Automatic Model Selection with Large Language Models for Reasoning                                          |      |        | 2023 | self-consistency<br>majority voting |
| 4    | GPT-4<br>(PHP, SC K=40)                                     | 96.5      |                         | ×                         | Progressive-Hint Prompting Improves Reasoning in Large Language Models                                      |      |        | 2023 | self-consistency<br>majority voting |
| 5    | GPT-4<br>(Model Selection, SC K=5)                          | 96.5      |                         | ×                         | Automatic Model Selection with Large Language Models for Reasoning                                          |      |        | 2023 | self-consistency<br>majority voting |
| 6    | SFT-Mistral-7B<br>(Metamath, OVM, Smart Ensemble)           | 96.4      | 7                       | ✓                         |                                                                                                             |      |        | 2024 |                                     |
| 7    | SFT-Mistral-7B<br>(AugData + ovm + ensemble)                | 95.9      | 7                       | ✓                         |                                                                                                             |      |        | 2024 |                                     |
| 8    | GPT-4<br>(PHP)                                              | 95.5      |                         | ×                         | Progressive-Hint Prompting Improves Reasoning in Large Language Models                                      |      |        | 2023 |                                     |
| 9    | MindOpt Copilot Mistral-7B<br>(MetaMath, OVM, BS, Ensemble) | 95.1      | 7                       | ✓                         |                                                                                                             |      |        | 2024 |                                     |
| 10   | Claude 3 Opus<br>(0-shot chain-of-thought)                  | 95        |                         | ×                         | The Claude 3 Model Family: Opus, Sonnet, Haiku                                                              |      |        | 2024 | chain-of-thought<br>zero-shot       |
| 11   | Gemini Ultra<br>(maj1@32)                                   | 94.4      |                         | ×                         | Gemini: A Family of Highly Capable Multimodal Models                                                        |      |        | 2023 | majority voting                     |
| 12   | SFT-Mistral-7B<br>(Metamath + ovm + ensemble)               | 94.13     | 7                       | ✓                         |                                                                                                             |      |        | 2024 |                                     |

<https://paperswithcode.com/sota/arithmetic-reasoning-on-gsm8k>

Fei Yu, Anningzhe Gao, **Benyou Wang**. OVM, Outcome-supervised Value Models for Planning in Mathematical Reasoning. <https://arxiv.org/abs/2311.13951>. Findings of NAACL 2024.

## II. AI for Math: o1-like reasoning for AIMO



<https://www.kaggle.com/c/ai-mathematical-olympiad-progress-prize-2>





# III. Multimodal LLMs

Data:

- **Allava**, a popular image-text dataset
- **Silkie**, first RLHF in multi-modal LLMs

Benchmark

- **MileBench**, long context benchmark for multi-modal LLMs
- **S2S-Arena**, First S2S benchmark
- **GMAI-benchmark**, comprehensive benchmark for medical imaging
- **Roadmap**, a leveled benchmark for speech understanding
- **AV-Odyssey Bench**, audio-visual benchmark

Models

- **LongLLaVA**, first multi-modal LLMs that process 1000+ images
- **HuatuogPT-vision**, 200,000 model downloading per month
- Speech **understading** LLM Soundwave
- End2end Speech LLMs **Echo-X**
- LLM-based image generation **Janus**

[1] Guiming Hardy Chen, Shunian Chen, Ruifei Zhang, Junying Chen, Xiangbo Wu, Zhiyi Zhang, Zhihong Chen, Jianquan Li, Xiang Wan, **Benyou Wang**. ALLaVA: Harnessing GPT4V-synthesized Data for A Lite Vision-Language Model. <https://arxiv.org/abs/2402.11684>

[2] D Song, S Chen, GH Chen, F Yu, X Wan, **B Wang**. Milebench: Benchmarking mllms in long context, arXiv preprint arXiv:2404.18532. **COLM 2024**

[3] Ling-Hao Chen, Shunlin Lu, Ailing Zeng, Hao Zhang, Benyou Wang, Ruimao Zhang, Lei Zhang. MotionLLM: Understanding Human Behaviors from Human Motions and Videos. <https://arxiv.org/abs/2405.20340>

[4] Wentao Ge, Shunian Chen, Guiming Chen, Junying Chen, Zhihong Chen, Shuo Yan, Chenghao Zhu, Ziyue Lin, Wenya Xie, Xidong Wang, Anningzhe Gao, Zhiyi Zhang, Jianquan Li, Xiang Wan, Benyou Wang. Mllm-bench, evaluating multi-modal llms using gpt-4v. <https://arxiv.org/abs/2311.13951>

[5] Lei Li, Zhihui Xie, Mukai Li, Shunian Chen, Peiyi Wang, Liang Chen, Yazheng Yang, Benyou Wang, Lingpeng Kong. Silk: Preference distillation for large visual language models. <https://arxiv.org/abs/2312.10665>. **EMNLP 2024**

[6] Zhihong Chen, Shizhe Diao, Benyou Wang, Guanbin Li, and Xiang Wan. "Towards Unifying Medical Vision-and-Language Pre-training via Soft Prompts". ICCV 2023

[7] Pengcheng Chen, Jin Ye, Guoan Wang, Yanjun Li, Zhongying Deng, Wei Li, Tianbin Li, Haodong Duan, Ziyang Huang, Yanzhou Su, Benyou Wang, Shaoting Zhang, Bin Fu, Jianfei Cai, Bohan Zhuang, Eric J Seibel, Junjun He, Yu Qiao. GMAI-MM Bench: A Comprehensive Multimodal Evaluation Benchmark Towards General Medical AI. **NeurIPS 2024 Track Datasets and Benchmarks**

[8] Xidong Wang, Dingjie Song, Shunian Chen, Chen Zhang, Benyou Wang. LongLLaVA: Scaling Multi-modal LLMs to 1000 Images Efficiently via a Hybrid Architecture. <https://arxiv.org/abs/2409.02889>

# Soundwave, A Data-efficient speech LLM

## Highlights

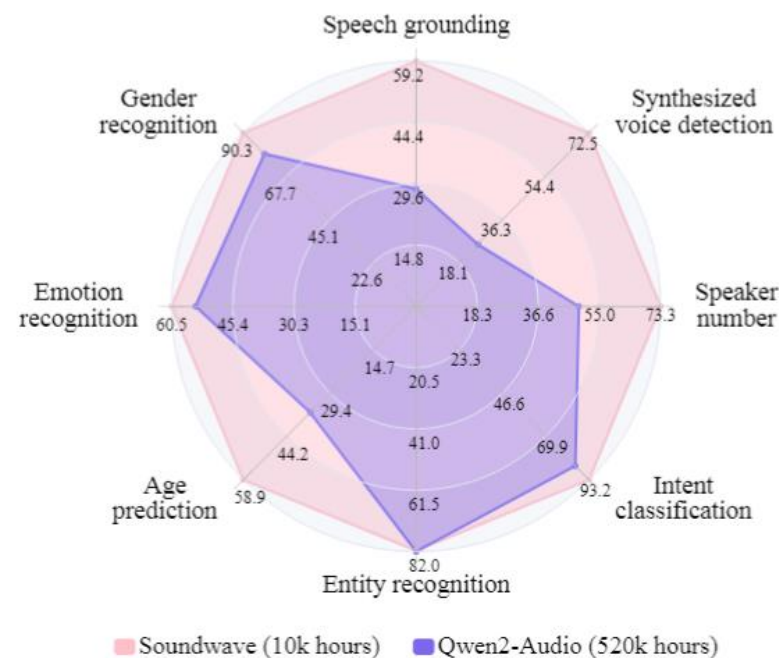
- Less data: 1/50 that of Qwen2-Audio
- Better performance than Qwen2-audio

One could **achieve it with an academic budget**

How:

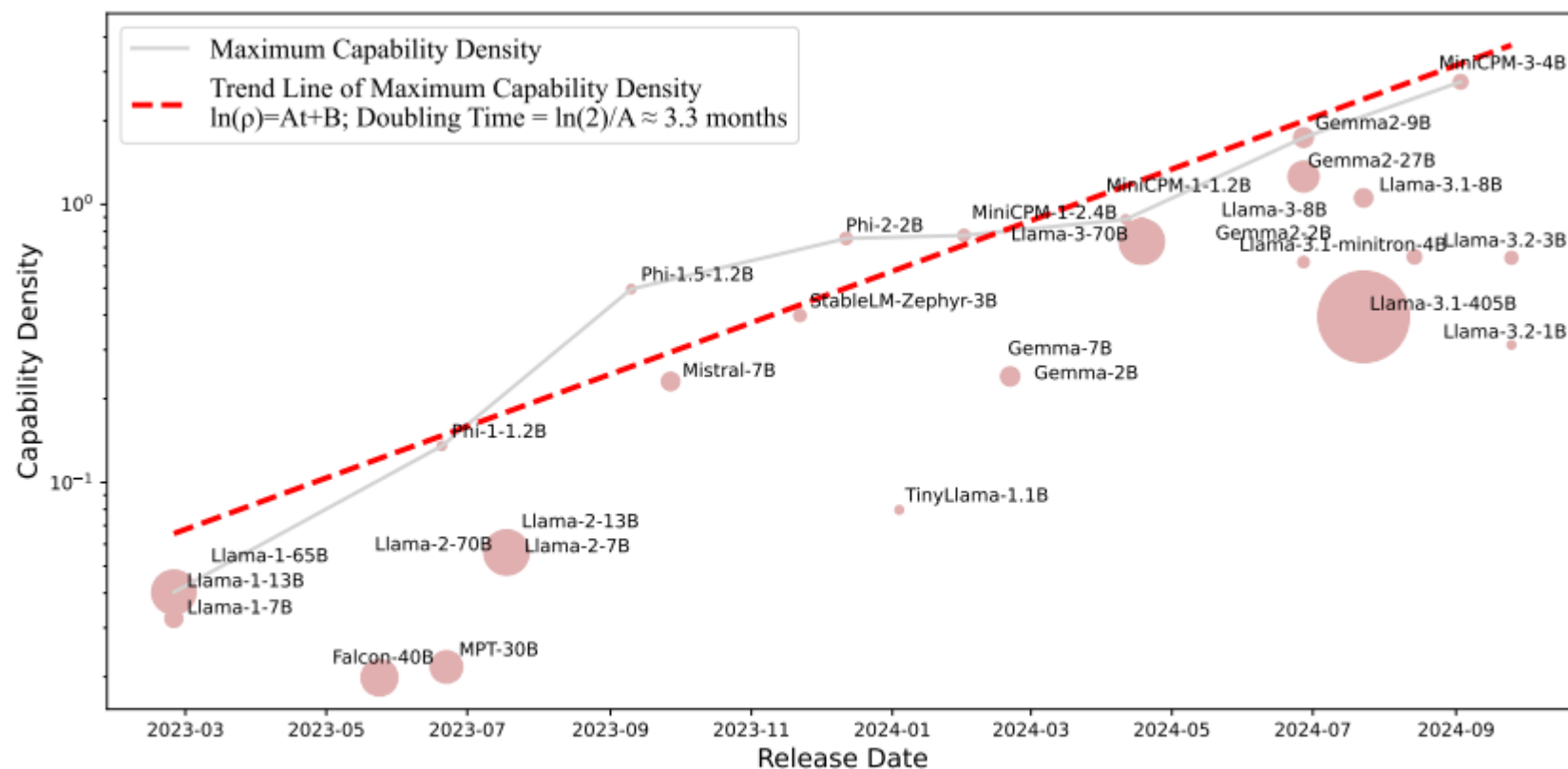
Additional CTC loss for better speech-text alignment

Better data engineering



Yuhao Zhang, et.al. **Benyou Wang**. Soundwave : Less is More for Speech-text Alignment in LLMs.

# IV. Cheaper deployment



**Figure 1:** The estimated capability density of open-source base LLMs.

Models become denser (performance/size)



# Impact: ranked 14<sup>th</sup> worldwide

Spaces osanseviero universities like 2 Running

App Files Community

## Universities at Hugging Face

This is from November 10 2023, it's not automatically updated.

| total_likes | name                                  | count_users | models_count | models_likes | models_downloads | datasets_count | datasets_likes | datasets_downloads |
|-------------|---------------------------------------|-------------|--------------|--------------|------------------|----------------|----------------|--------------------|
| 11254       | <a href="#">CompVis</a>               | 23          | 13           | 9884         | 4611345          | 0              | 0              | 0                  |
| 8224        | <a href="#">THUDM</a>                 | 29          | 29           | 7034         | 757300           | 5              | 236            | 15582              |
| 3427        | <a href="#">sentence-transformers</a> | 4           | 124          | 3276         | 20915487         | 5              | 76             | 89                 |
| 1330        | <a href="#">Helsinki-NLP</a>          | 14          | 1440         | 1290         | 5018692          | 1              | 40             | 63329              |
| 968         | <a href="#">shi-labs</a>              | 6           | 21           | 85           | 106372           | 1              | 0              | 27                 |
| 862         | <a href="#">hkunlp</a>                | 14          | 55           | 831          | 327200           | 0              | 0              | 0                  |
| 700         | <a href="#">fnlp</a>                  | 11          | 30           | 570          | 5027             | 4              | 130            | 43                 |
| 560         | <a href="#">tatsu-lab</a>             | 4           | 12           | 44           | 159              | 3              | 516            | 80701              |
| 498         | <a href="#">uanlp</a>                 | 9           | 1            | 7            | 0                | 0              | 0              | 0                  |
| 466         | <a href="#">declare-lab</a>           | 13          | 16           | 362          | 96854            | 7              | 34             | 149                |
| 455         | <a href="#">weizmannscience</a>       | 9           | 0            | 0            | 0                | 0              | 0              | 0                  |
| 430         | <a href="#">stanfordnlp</a>           | 10          | 105          | 181          | 55422            | 1              | 239            | 5534               |
| 403         | <a href="#">MBZUAI</a>                | 68          | 27           | 254          | 125298           | 4              | 149            | 12112              |
| 339         | <a href="#">FreedomIntelligence</a>   | 23          | 21           | 129          | 11671            | 59             | 210            | 1816               |
| 331         | <a href="#">csebuettlp</a>            | 7           | 20           | 256          | 667251           | 7              | 75             | 34044              |
| 329         | <a href="#">poloclub</a>              | 6           | 2            | 2            | 0                | 1              | 327            | 1370149            |
| 298         | <a href="#">cvlab</a>                 | 4           | 1            | 9            | 0                | 0              | 0              | 0                  |

Ranked 14<sup>th</sup> in worldwide University ranking in Huggingface

# Impact: 10,000 GitHub Stars

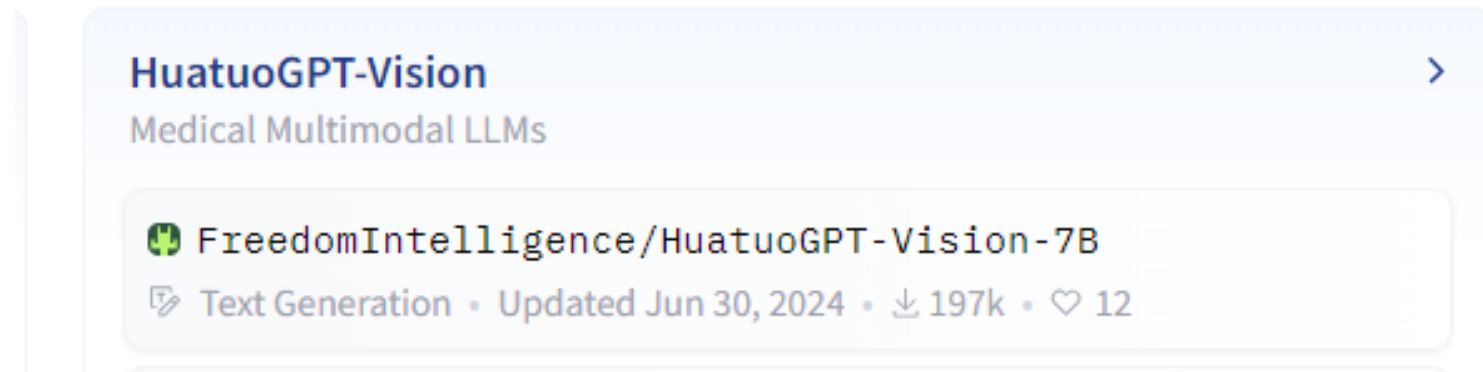
All New repository

sort: [stars](#)

| 167 repositories                                                                                                                                                                                                                                               |        | Stars |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|------|
|  <b>LLMZoo</b> ⚡ LLM Zoo is a project that provides data, models, and evaluation benchmark for large language models. ⚡                                                       | Python | 201   | 2.9k |
|  <b>Medical_NLP</b> Medical NLP Competition, dataset, large models, paper                                                                                                     |        | 411   | 2.2k |
|  <b>HuatuoGPT</b> HuatuoGPT, Towards Taming Language Models To Be a Doctor. (An Open Medical GPT)                                                                             | Python | 146   | 1.1k |
|  <b>TextClassificationBenchmark</b> A Benchmark of Text Classification in PyTorch                                                                                             | Python | 137   | 602  |
|  <b>HuatuoGPT-o1</b> Medical o1, Towards medical complex reasoning with LLMs                                                                                                  | Python | 42    | 546  |
|  <b>HuatuoGPT-II</b> HuatuoGPT2, One-stage Training for Medical Adaption of LLMs. (An Open Medical GPT)                                                                       | Python | 61    | 385  |
|  <b>InstructionZoo</b>                                                                                                                                                        |        | 24    | 276  |
|  <b>HuatuoGPT-Vision</b> Medical Multimodal LLMs                                                                                                                              | Python | 23    | 270  |
|  <b>ALLaVA</b> Harnessing 1.4M GPT4V-synthesized Data for A Lite Vision-Language Model                                                                                        | Python | 9     | 251  |
|  <b>Huatuo-26M</b> The Largest-scale Chinese Medical QA Dataset: with 26,000,000 question answer pairs.                                                                       |        | 24    | 231  |
|  <b>LongLLaVA</b> LongLLaVA: Scaling Multi-modal LLMs to 1000 Images Efficiently via Hybrid Architecture                                                                      | Python | 13    | 187  |
|  <b>ReasoningNLP</b> paper list on reasoning in NLP                                                                                                                           |        | 15    | 183  |
|  <b>Apollo</b> Multilingual Medicine: Model, Dataset, Benchmark, Code                                                                                                         | Python | 8     | 178  |
|  <b>crosstalk-generation</b> Code and data for crosstalk text generation tasks, exploring whether large models and pre-trained language models can understand humor.         | Python | 17    | 168  |
|  <b>GrammarGPT</b> The code and data for GrammarGPT.                                                                                                                        | Python | 9     | 167  |
|  <b>CMB</b> CMB, A Comprehensive Medical Benchmark in Chinese                                                                                                               | Python | 15    | 148  |
|  <b>Evaluation-of-ChatGPT-on-Information-Extraction</b> An Evaluation of ChatGPT on Information Extraction task, including Named Entity Recognition (NER), Relation Extr... | Python | 11    | 130  |
|  <b>AceGPT</b>                                                                                                                                                              | Python | 7     | 116  |

Received more than **10,000** stars in total

Impact: **200,000** model downloading per month



**200,000** model downloading in a single month in HuggingFace

Thanks

Join us as a RA if interested  
[wangbenyou@cuhk.edu.cn](mailto:wangbenyou@cuhk.edu.cn)